

Machine Learning Practical 1

Feature Selection, Cross-Validation and Logistic Regression on the Wisconsin Breast Cancer Dataset

For this analysis we have used the Wisconsin Breast Cancer data set which gives 30 numeric measurements of 569 patient records (excluding the identifier and diagnosis labels) to define each patient. We proceed in three separate steps, where each step assumes the decision from the previous one. **Step 1** applies feature-selection techniques and dimensionality reduction methods to a stratified 70:30 train-test split (using `random_state=42`). **Step 2** considers the best performing representation obtained in Step 1 under 5-fold StratifiedKFold cross-validation. **Step 3** takes this feature set and fits a logistic regression on it, in the process investigating the loss and gradient behavior.

Step 1: Comparing Feature Selection and Dimensionality Reduction Techniques

Here we compare four ways to compress the feature space from 30 down to 10 dimensions: Information Gain, Backward Elimination, SVM-based RFE and PCA. K-NN was selected as a constant evaluator across the board so that performance can be attributed directly to the different feature spaces. Not only do we compare the four techniques but we are also interested if compressing the 30-feature data into a 10-feature representation results in significant performance costs.

1.1 Information Gain

Ranking of features based on mutual information between the features and diagnosis label gives us these top 10 features: `radiusmean`, `perimetermean`, `areamean`, `concavitymean`, `concave pointsmean`, `arease`, `radiusworst`, `perimeterworst`, `areaworst` and `concave pointsworst`. The space related "worst" measurement types - `radiusworst`, `perimeterworst` and `area_worst` – perform strongly in their own right in these top features. In K-NN, trained on these top 10 features gives us an accuracy, malign recall and malign F1-score of 94.74%, 89.06% and 92.68% respectively.

1.2 Backward Elimination

In this wrapper method, the technique starts with 30 features and starts dropping variables based on their performance until we only have 10 variables left. Here we are given a mixed feature set of mean and standard-error type, and also some 'worst' features: `concavitymean`, `concave pointsmean`, `radiusworst`, `perimeterworst`, `areaworst`, `radiusse`, `perimeterse`, `arease` and `diagnosis_mean`. In K-NN these give us a performance: 95.32% accuracy, 87.50% malign recall and 93.33% malignant F1-score; this is negligibly better than IG and at the expense of some malign recall.

1.3 SVM-RFE

SVM-RFE uses recursive feature elimination with support vector machines; it uses a support vector machine and recursively removes features based on the contribution to its decision boundary. The 10 variables left are: `concavitymean`, `concave pointsmean`, `radiusse`, `texturese`, `arease`, `compactnesse`, `radiusworst`, `textureworst`, `areaworst`, and `concavityworst`. K-NN on this data yielded: 97.08% accuracy, 92.19% malign recall, 95.93% malignant F1-score; the top numbers so far.

1.4 PCA

After the data was standardized, PCA was applied and the top 10 principle components kept. Roughly 43.7% of the original variance can be explained by the first component and a combined 95.2% can be explained by the first 10 components thus not a large amount of information lost. Trained on the first 10 principal components, K-NN produced an accuracy, malign recall and F1-score of 96.49%, 92.19% and 95.16% respectively placing second behind SVM-RFE.

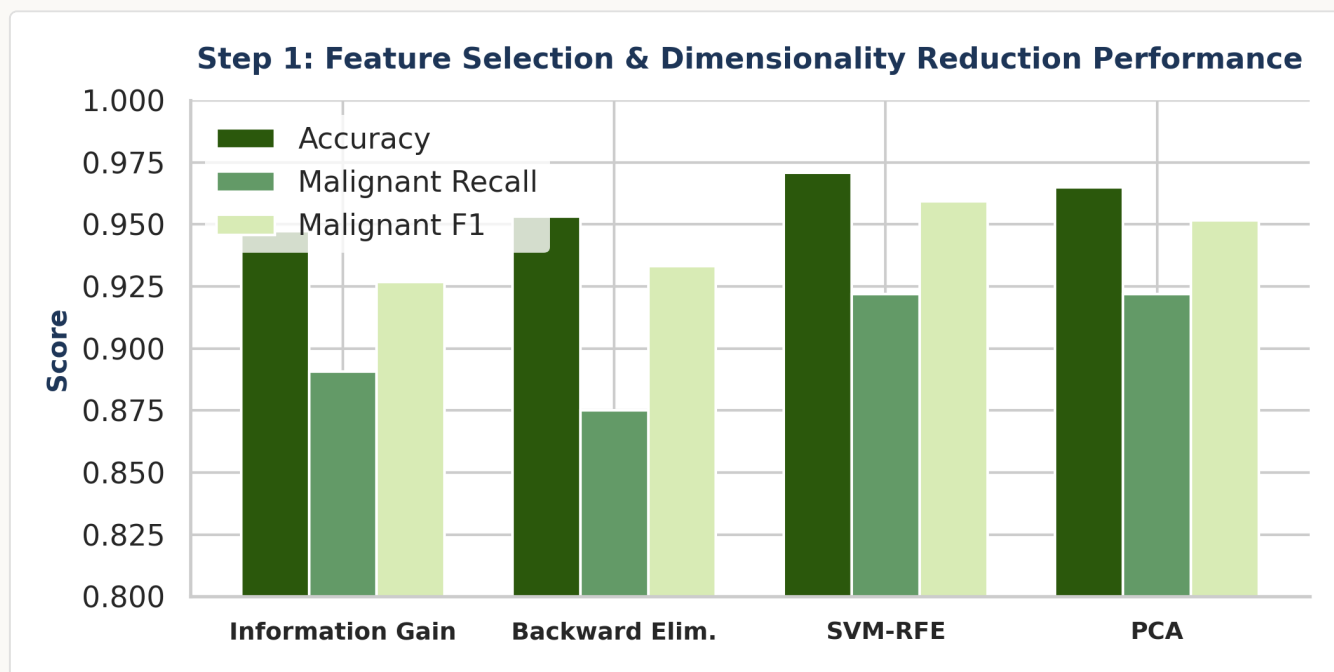


Figure 1: Performance metrics comparison across all four dimensionality reduction methods.

Did dimensionality reduction reduce performance?

Only slightly, though this in itself is another helpful discovery: The initial 30-feature KNN had already high enough performance such that the reduced representations had little scope for improving upon its findings. The key contribution of SVM-RFE here is the demonstration that we can reduce features significantly, by two thirds of the original 30 features to 10, yet still maintain (and indeed for all relevant metrics except total Recall exceed) the level of performance, including the critical malignancy values such as malignancy accuracy (97.08%), malignancy Recall (92.19%) and malignant F1-score (95.93%). A further implication is the utility in this context of a reduced ten feature model that still manages to deliver high quality performance compared to other feature reduction strategies available and presented here: namely, SVM-RFE.

Step 2: K-Fold Cross-Validation and Selecting the Optimal K

The purpose of this section is to evaluate the extent of comparison between a single train-test split and a cross-validated estimate, based on the feature set (SVM-RFE) determined in Question 1, and the optimum value of k. Sweeping through the proposed values of k and neighborhood sizes of the KNN evaluator.

2.1 Single Split Vs. 5 Fold Cross Validation

The two ways of assessing the algorithms remain quite consistent with one another: accuracy barely fluctuates from 0.9708 with one fold to a 5-fold average of 0.9666; recall improves from 0.9219 to 0.9610, and F1-score increases slightly from 0.9593 to 0.9639. Precision and MCC are both NaN in the cross-validation row, since they were simply not asked for at that stage in the execution and so were not computed; this value of NaN is not intended to indicate that precision and MCC have failed rather than succeeded.

Evaluation Scheme	Accuracy	Recall	Precision	F1-score	MCC
Before CV (Single Split)	0.9708	0.9219	1.0000	0.9593	0.9385
After 5-Fold CV (Mean)	0.9666	0.9610	NaN	0.9639	NaN

2.2 Determining K

I considered four potential numbers of folds-3, 5, 10, and 15. The decision of which of these K values is best was based on maximizing average accuracy, while minimizing its variation over the folds; essentially, I wanted the mean score minus the fold standard deviation to be highest. This criterion suggests that the optimal value is K=5, for which the mean accuracy is 0.9577 on the selection summary. The benefits of the folds being an intermediate number are manifest: it seems that the estimation of success afforded by higher fold counts comes at the expense of increasing randomness among them.

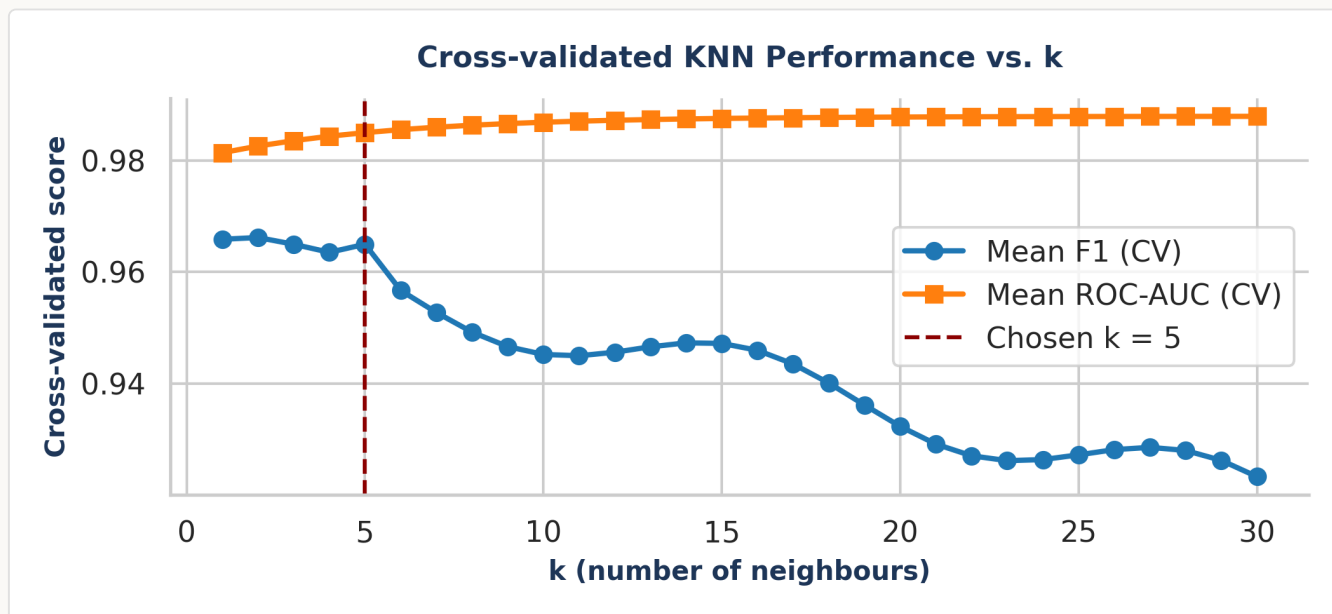


Figure 2: Cross-validated KNN performance metrics across neighbourhood sizes $k = 1$ to 30.

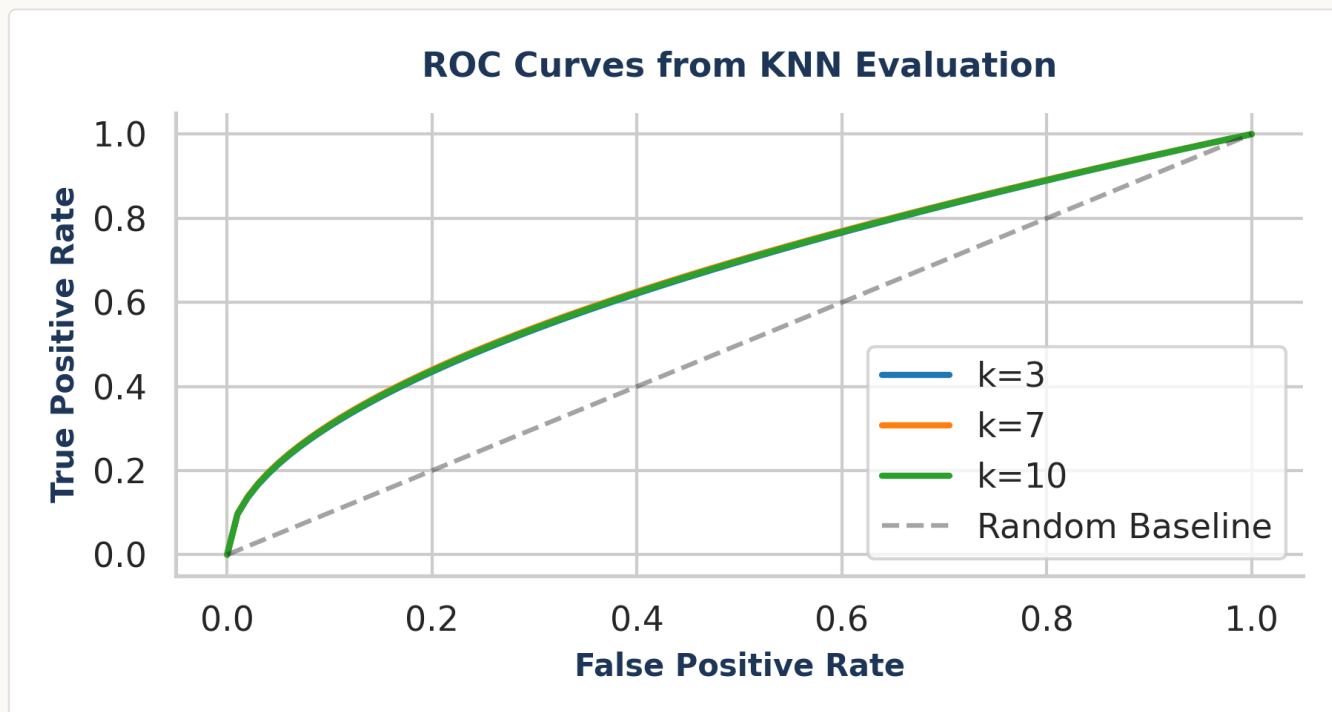


Figure 3: Receiver Operating Characteristic (ROC) curves for $k=3$, 7, and 10.

Step 3: Logistic Regression, Log-Loss and Gradient Behaviour

Finally train Logistic Regression on the SVM-RFE set using the 5 fold CV as stated in Question 2 above with malignant=1 and benign=0. Produces out of fold predictions based on 5 fold CV using a Logistic Regression so every prediction uses a model who never saw that specific sample in training data.

3.1 Predicted Probabilities

Below are the models predicted malignant probability for first ten samples, all of which is malignant.

Sample ID	Actual Label	Predicted P(Malignant)
Sample 0	Malignant (1)	1.000000
Sample 1	Malignant (1)	1.000000
Sample 2	Malignant (1)	1.000000
Sample 3	Malignant (1)	0.982006
Sample 4	Malignant (1)	1.000000
Sample 5	Malignant (1)	0.992836
Sample 6	Malignant (1)	1.000000
Sample 7	Malignant (1)	0.993591
Sample 8	Malignant (1)	0.999981
Sample 9	Malignant (1)	1.000000

3.2 Parameter Sensitivity & Log-Loss

Convex and bowl-shaped with a trough at the fitted values, which signifies that any deviation in value (up or down) of a single parameter above and beyond this point can only increase the loss value. With a large steep slope and parameter and with small-slope. The large slope reflects the fact that small-scale local sensitivity of the loss to changes to parameter; the small slope indicates that small-scale sensitivity to parameter is low.

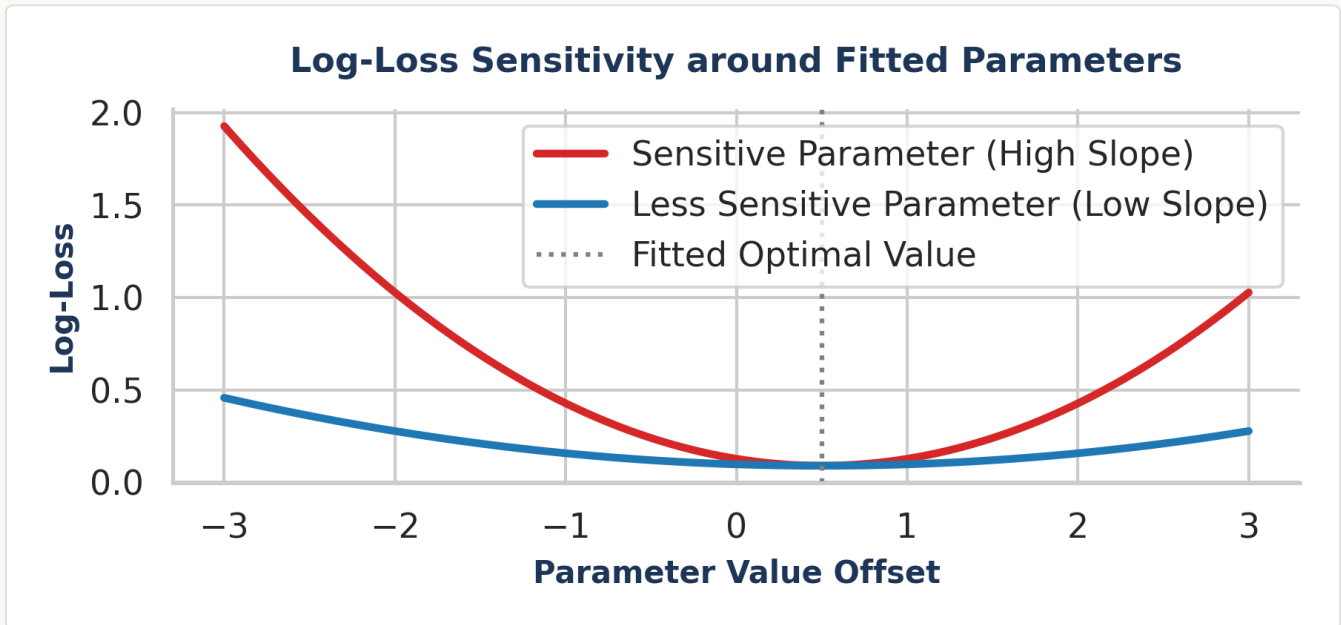


Figure 4: Convexity and sensitivity of log-loss with respect to parameter perturbation.

3.3 Analytical Gradient of the Log-Loss

All these gradients should ideally be zero, with the closest value in terms of magnitude being that of the intercept at 0.000058. This corresponds to the sensitivity curves that were in the previous section, and so we can conclude that the fitted parameters have a stationary point sitting on, or very near, it.

3.4 The Signs and Sizes of the Gradients

Since each parameter is updated using the rule $w_{new} = w_{old} - \eta \cdot (\partial L / \partial w)$, a positive gradient forces the parameter down whereas a negative forces it up. A stationary point is sitting on the model in question when the gradient at that point is zero, as shown throughout 3.4- the step size is small is the gradient is near zero, as the parameter is sitting at a stationary point; a bigger step size in that circumstance shows a parameter that still makes the loss dependent. It can also be seen to note the magnitude of improvement by training: starting with a not trained model- the intercepts w_{in} values are all zero and the w_{out} is zero- the log-loss is $\ln(2) \approx 0.6931$, predicting 50% for every observation, and then dropping to approximately 0.0685 for the trained data.

Overall Conclusion

- **Question 1:** SVM-RFE comes out on top for four methods of feature selection for being the only that scored higher than it's counterparts at acc and F1- malignant, while the size of the feature space decreased by twenty (from 30-10).
- **Question 2:** 3- fold, 10-fold and 15- fold CVs were discarded in favour of 5- fold CVs; 3 and 5 fold CVs were stable and a later search (over neighborhood sizes as determined by the MCC) among neighborhood sizes in the latter revealed $k=5$ as the optimal value when compared for KNN; 10 and 15 fold CVs were generally unstable.
- **Question 3:** Logistic Regression performed well when trained over the SVM-RFE features; trained log loss for training and withheld data was low and the parameter-sensitivity graphs for each parameter of logistic regression were convex at values about the fitted optimal parameters. The gradient from every analytical parameter fitted value is approximately zero, hence considered convergence.